The energy of an electron in the nth orbit

Since the electrostatic force is a conservative force, the potential energy for the nth orbit is

$$U_n = \frac{1}{4\pi\varepsilon_0}\frac{(+Ze)(-e)}{r_n} = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n}$$

$$= -\frac{1}{4\varepsilon_0^2}\frac{Z^2me^4}{h^2n^2} \quad \left(\because r_n = \frac{\varepsilon_0 h^2}{\pi me^2}\frac{n^2}{Z}\right)$$

The kinetic energy of the electron in n^{th} orbit is

$$KE_n = \frac{1}{2}m\upsilon_n^2 = \frac{me^4}{8\varepsilon_0^2h^2}\frac{Z^2}{n^2}$$

This implies that $U_n = -2\ KE_n$. Total energy of the electron in in the nth orbit is

$$E_n = KE_n + U_n = KE_n - 2KE_n = -KE_n$$

$$E_n = -\frac{me^4}{8\varepsilon_0^2h^2}\frac{Z^2}{n^2}$$

For hydrogen atom (Z = 1),

$$E_n = -\frac{me^4}{8\varepsilon_0^2h^2}\frac{1}{n^2}\ \text{joule} \qquad (9.17)$$

where n stands for principal quantum number. The negative sign in equation (9.17) indicates that the electron is bound to the nucleus.

Substituting the values of mass and charge of an electron (m and e), permittivity of free space ε_0 and Planck's constant h and expressing energy in terms of electron(+(eV)), we get

$$E_n = -13.6\frac{1}{n^2}\ eV$$

For the first orbit (ground state), the total energy of electron is $E_1 = -13.6\ eV$.
For the second orbit (first excited state), the total energy of electron is $E_2 = -3.4\ eV$.
For the third orbit (second excited state), the total energy of electron is $E_3 = -1.51\ eV$ and so on.

Notice that the energy of the first excited state is greater than that of the ground state, second excited state is greater than that of the first excited state and so on. Thus, the orbit which is closest to the nucleus (r_1) has lowest energy (minimum energy what it is compared with other orbits). So, it is often called ground state energy (lowest energy state). The ground state energy of hydrogen (–13.6 eV) is used as a unit of energy called Rydberg (1 Rydberg = –13.6 eV).

The negative value of this energy is because of the way the zero of the potential energy is defined. When the electron is taken away to an infinite distance (very far distance) from nucleus, both the potential energy and kinetic energy terms vanish and hence the total energy also vanishes.

The energy level diagram along with the shape of the orbits for increasing values of n are shown in Figure 9.20. It shows that the energies of the excited states come closer and closer together when the principal quantum number n takes higher values.

EXAMPLE 9.3

(a) Show that the ratio of velocity of an electron in the first Bohr orbit to the speed of light c is a dimensionless number.

(b) Compute the velocity of electrons in ground state, first excited state and second excited state in Bohr atom model for hydrogen atom.

Solution

(a) The velocity of an electron in n^{th} orbit is

$$\upsilon_n = \frac{h}{2\pi ma_0}\frac{Z}{n}$$

where $a_0 = \frac{\varepsilon_0 h^2}{\pi me^2}$ = Bohr radius. Substituting for a_0 in υ_n,

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